

Matgeo Presentation - Problem 4.13.58

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Problem Statement

Question 4.13.58

Show that all chords of the curve $3x^2 - y^2 - 2x + 4y = 0$, which subtend a right angle at the origin, pass through a fixed point. Find the coordinates of the point.

Solution: Matrix Representation

The equation of the given curve is $3x^2 - y^2 - 2x + 4y = 0$.

Let $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. The curve in matrix form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ is:

$$\mathbf{x}^\top \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}^\top \mathbf{x} + 0 = 0. \quad (0.1)$$

Here, the matrix of the quadratic part is

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \quad (0.2)$$

the vector for the linear part is

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad (0.3)$$

and the constant term is $f = 0$.

Solution: Homogenization

Let the equation of a chord PQ of the curve be $lx + my = 1$.

In vector form, this is:

$$\mathbf{n}^\top \mathbf{x} = 1, \text{ where } \mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \quad (0.4)$$

The combined equation of the lines from the origin to the intersection points is found by homogenizing equation (0.1) using equation (0.4).

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2(\mathbf{u}^\top \mathbf{x})(1) = 0 \quad (0.5)$$

Substituting $\mathbf{n}^\top \mathbf{x}$ for 1:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2(\mathbf{u}^\top \mathbf{x})(\mathbf{n}^\top \mathbf{x}) = 0 \quad (0.6)$$

Solution: Pair of Lines

This can be written as a single quadratic form:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + \mathbf{x}^T (\mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T) \mathbf{x} = 0 \quad (0.7)$$

$$\mathbf{x}^T (\mathbf{V} + \mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T) \mathbf{x} = 0 \quad (0.8)$$

This is the equation of the pair of lines OP and OQ. Let

$$W = \mathbf{V} + \mathbf{u} \mathbf{n}^T + \mathbf{n} \mathbf{u}^T. \quad (0.9)$$

The lines are perpendicular if and only if the trace of the matrix W is zero.

Justification: Trace and Perpendicularity

- 1) **Factoring the Quadratic Form:** The equation $\mathbf{x}^\top \mathbf{W} \mathbf{x} = 0$ represents two lines passing through the origin. If their normal vectors are $\mathbf{n}_1$ and $\mathbf{n}_2$, the equation is $(\mathbf{n}_1^\top \mathbf{x})(\mathbf{n}_2^\top \mathbf{x}) = 0$. This means $\mathbf{W} = k(\mathbf{n}_1 \mathbf{n}_2^\top + \mathbf{n}_2 \mathbf{n}_1^\top)$.
- 2) **Condition for Perpendicular Lines:** Two lines are perpendicular if their normal vectors are orthogonal, i.e., $\mathbf{n}_1^\top \mathbf{n}_2 = 0$.
- 3) **Connecting to the Trace:**

$$\begin{aligned}\text{trace}(\mathbf{W}) &= \text{trace}\left(k(\mathbf{n}_1 \mathbf{n}_2^\top + \mathbf{n}_2 \mathbf{n}_1^\top)\right) \\ &= k\left(\mathbf{n}_2^\top \mathbf{n}_1 + \mathbf{n}_1^\top \mathbf{n}_2\right) = 2k(\mathbf{n}_1^\top \mathbf{n}_2)\end{aligned}$$

Therefore, $\text{trace}(\mathbf{W}) = 0 \iff \mathbf{n}_1^\top \mathbf{n}_2 = 0$.

Solution: Applying the Trace Condition

The condition for the lines to be perpendicular is $\text{trace}(W) = 0$.

$$\text{trace}(W) = \text{trace}(\mathbf{V}) + \text{trace}(\mathbf{u}\mathbf{n}^\top) + \text{trace}(\mathbf{n}\mathbf{u}^\top) \quad (0.13)$$

$$\text{trace}(W) = \text{trace}(\mathbf{V}) + \mathbf{n}^\top \mathbf{u} + \mathbf{u}^\top \mathbf{n} = \text{trace}(\mathbf{V}) + 2\mathbf{u}^\top \mathbf{n} \quad (0.14)$$

The condition for the chord to subtend a right angle at the origin is

$$\text{trace}(\mathbf{V}) + 2\mathbf{u}^\top \mathbf{n} = 0 \quad (0.17)$$

Solution: Deriving the Condition

From the problem statement:

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \implies \text{trace}(\mathbf{V}) = 3 + (-1) = 2 \quad (0.18)$$

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad (0.20)$$

$$\mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \quad (0.21)$$

Substituting these values into equation (0.17):

$$2 + 2 \begin{pmatrix} -1 & 2 \end{pmatrix} \begin{pmatrix} l \\ m \end{pmatrix} = 0 \quad (0.22)$$

$$2 + 2(-l + 2m) = 0 \quad (0.1)$$

$$l - 2m = 1 \quad (0.23)$$

Solution: Finding the Fixed Point

This is the condition on the coefficients of the chord $lx + my = 1$. We want to show this line passes through a fixed point (x_0, y_0) .

$$lx_0 + my_0 = 1 \quad (0.24)$$

From (0.23), $l = 1 + 2m$. Substituting this into (0.24):

$$(1 + 2m)x_0 + my_0 = 1 \quad (0.25)$$

$$(x_0 - 1) + m(2x_0 + y_0) = 0 \quad (0.27)$$

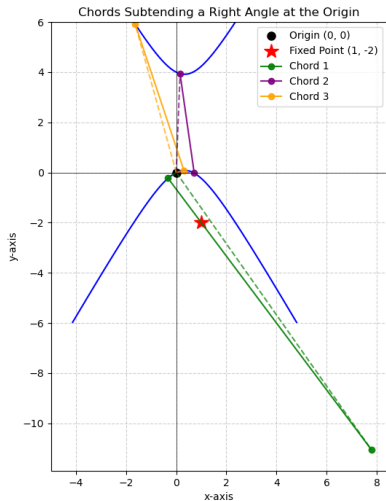
This must hold for all m , so the coefficients must be zero.

$$x_0 - 1 = 0 \implies x_0 = 1 \quad (0.28)$$

$$2x_0 + y_0 = 0 \implies 2(1) + y_0 = 0 \implies y_0 = -2 \quad (0.30)$$

Therefore, all such chords pass through the fixed point $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

Plot



C Code: solver.c

```
#include <math.h>
#include <stdio.h>

int find_intersections(double l, double m, double* out_points) {
    double a, b, c;

    if (fabs(m) < 1e-9) { // Vertical line case
        double x = 1.0 / l;
        a = 1.0; b = -4.0; c = -(3.0/(1*1) - 2.0/l);
        double disc = b*b - 4*a*c;
        if (disc < 0) return -1;
        out_points[0] = x; out_points[2] = x;
        out_points[1] = (-b + sqrt(disc)) / (2*a);
        out_points[3] = (-b - sqrt(disc)) / (2*a);
        return 0;
    }

    a = 3 * m * m - 1 * l;
    b = 2 * l - 2 * m * m - 4 * l * m;
    c = 4 * m - 1;
```

Python Code: visualizer.py

```
import ctypes, os
import numpy as np
import matplotlib.pyplot as plt

lib_name = "./libsolver.so" # Adapt for Windows/macOS if needed
c_lib = ctypes.CDLL(lib_name)
c_lib.find_intersections.argtypes = [ctypes.c_double, ctypes.c_double, ctypes.POINTER(ctypes.c_double)]
c_lib.find_intersections.restype = ctypes.c_int

fig, ax = plt.subplots(figsize=(10, 8))
x = np.linspace(-5, 7, 400); y = np.linspace(-6, 6, 400)
X, Y = np.meshgrid(x, y)
curve_eq = 3*X**2 - Y**2 - 2*X + 4*Y
ax.contour(X, Y, curve_eq, levels=[0], colors='blue')

fixed_point = (1, -2)
ax.plot(0, 0, 'ko', markersize=8, label='Origin')
ax.plot(fixed_point[0], fixed_point[1], 'r*', markersize=15,
        label=f'Fixed Point')
```